engineering 4 us all

Student Handout

Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element I Initial Template - Testing Results

1. Grip Tightly Onto the String (high)

Process:

Attach the prototype to a hoodie string

Hold the hoodie by the sleeves

Wring the hoodie around 5 times times

Results:

Grip Tightly Onto the String (high)	
Hoodie Design 1:	successfully stayed on completely
Hoodie Design 2:	slipped a little bit but overall did not fall off
Hoodie Design 3:	stayed on completely with some movement

Based on our testing data, we found that the prototype stays onto the different hoodie types each time that means the size of the holes on the prototype is appropriate because the string is able to hold tightly.

Friction might not always be enough to hold the device in place. Maybe think of another alternative using velcro or something more reliable

2. Sturdy Test (high)

Process:

Take the prototype and put it on the floor

Step directly onto the prototype (100-150lbs)

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Results:

Sturdy Test (high)	
Top:	completely sturdy, no scratches
Side:	cannot fully step onto the side, but saw some bending in the device
Back:	completely sturdy

Based on our testing data, we found that the agreeing results means the device is appropriately sturdy and strong because there was no visible breakage or damage to the device.

All good, but would be better if the real design was made out of a more rubbery material to prevent from clanking while in the drying and washing machines

3. Keep the String from getting lost(high)

Process:

Tie one end of the hoodie string to a 20lb weight

Put the device onto the other end of the hoodie string

Place hoodie on the table or somewhere with an edge

Keep the hoodie in place with the hood dangling off the edge and carefully drop the weight

Results:

Keep the String from getting lost (high)	
Hoodie Design 1:	Success, device kind of pulled on the hole a bit but overall helped

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Hoodie Design 2:	somewhat a success, device barely stayed on and string was so close to getting lost
Hoodie Design 3:	success, there was a little bit of slipping off the string, but overall worked

Based on our testing data, we found that the data was somewhat the same, which means the device is big enough to hold the string in place because there was little to no movement during our testing.

As mentioned above, would be best to use something more reliable other than friction to keep the string in place

4. Effortless Test(high)

Process:

Get a timer and have a person time you the time it takes to put the device on

Results:

Effortless Test (high)	
Trial 1:	5.5 sec
Trial 2:	5.7 sec
Trial 3:	5.4 sec
Trial 4:	6.4 sec
Average of all:	5.75 sec

Based on our testing data, we found that it takes about 5 seconds each time which means it is quick and easy to put on because it has reached our goal of putting on in less than 60 seconds.

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Although around 5-6 seconds is fast, might need to do something that will completely stay on in about 1-2 seconds. Possible something that clicks or snaps together instead of stringing the string through the holes.

Element I: Testing, Data Collection and Analysis						
	5	4	3	2	1	0
1. Quantity of tests conducted. The student conducts _____________tests for high priority requirements that are reasonable based on instructional context, or through physical or mathematical modeling.	*more than four*	*four*	*three*	*two*	*one*	*no*

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2. Importance of tests. The student conducts tests for____________ requirements that are reasonable based on instructional context, or through physical or mathematical modeling.	*all high priority*	*most high priority*	*half of high priority*	*some low and high priority*	*only non priority*	*no*
3. Comprehension of testing procedure. The student demonstrates ______________ understanding of testing procedure, including the gathering and analysis of resultant data.	*considerable*	*ample*	*adequate*	*partial or overly general*	*minimal*	*a failure of*
4. Detail of the design analysis. The analysis of the effectiveness with which the design met stated goals includes ______________ explanation (and summary) of the data.	*a consistently detailed (generously* supported by pictures, graphs, charts and other visuals*)*	*a generally detailed (generally* supported by pictures, graphs, charts and other visuals*)*	*a somewhat detailed (at least somewhat* supported by pictures, graphs, charts and other visuals*)*	*a partial (partially complete and/or partially correct* and *minimally* supported by pictures, graphs, and other visuals*)*	*an attempted (may not be* supported by pictures, graphs, charts and other visuals*)*	*no*